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INTERIM FEDERAL SPECIFICATION

MATS, FLOOR; DENTAL-CHAIR, RUBBER

This interim Federal specification was developed by the Defense Personnel Support Center, 2800 South 20th Street, Philadelphia, Pennsylvania 19101, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above. The General Services Administration has authorized the use of this interim Federal specification as a valid exception to Federal Specification.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers one type of rubber floor mat for use with dental chairs.

2. APPLICABLE SPECIFICATIONS, STANDARDS, DRAWINGS AND OTHER PUBLICATIONS

2.1 Specifications, standards and drawings. The following specifications and standards, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.

Military

MIL-L-10547	Liners, Case, and Sheet, Overwrap: Water-Vaporproof , or Waterproof, Flexible.

FSC 7220

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STANDARDS

Federal

FED. TEST METHOD STD. NO. 601 Rubber: Sampling and Testing

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington. D.C., 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D.C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

(Copies of Military specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction sample. Unless otherwise specified (see 6.1), prior to the start of production of the contract quantity, one preproduction sample of the floor mat shall be assembled, completely finished, and tested by the contractor. The sample shall be submitted, together with the contractor's inspection and test reports, to the contracting officer for examination and testing to determine compliance with the requirements of this specification.

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Written approval of the materials, construction, and performance of the sample shall be obtained from the contracting officer, or his authorized representative, prior to the manufacture and assembly of the remaining quantity to be manufactured. Further production of the item by the contractor, prior to this approval, shall be at the contractor's risk. Accepted preproduction sample shall become the property of the procuring activity and shall be included in the quantity called for in the contract or order. In the event that the preproduction sample is not approved, the contractor shall submit a new sample for approval, or modify the existent sample so as to remove the deficiencies cited.

3.2 Materials. The materials used in the fabrication of the floor mats shall be natural rubber, synthetic rubber, or a mixture of the two. The base of the mat shall be sponge rubber and the wearing surface, good quality rubber tile.

3.3 Construction. The rubber floor mat shall be semicircular in shape with a semicircular cutout. The mat shall be constructed in two layers, properly cemented or vulcanized together. The rubber tile surface shall be smooth, easily cleaned, and not adversely affected by floor waxes. The edges of the mat, with the exception of the cutout , shall be beveled approximately one inch. The underneath surface of the mat shall be treated to prevent sliding or creeping when used on a smooth floor.

3.4 Flexibility. When tested as specified in 4.4.1, the mat shall show no signs of breaking, cracking or weakness.

3.5 Friction. When tested as specified in 4.4.2, the adhesion of the tile surface to the sponge rubber base shall be such that the rate of separation shall be not more than 1 inch per minute.

3.6 Modulus. When tested as specified in 4.4.3, the stress at 10 percent elongation shall be not less than 125 pounds per square inch.

3.7 Dimensions. The mat shall be semicircular in shape with an overall size of not less than 69 inches by 35 inches. The mat shall have a semicircular cutout, with a diameter of 24 inches, enabling the mat to be placed around the base of a dental chair. The distance from the cutout to the outer edge of the mat shall be 23-1/2 inches plus or minus 1/2 inch. The layer of tile shall have a minimum thickness of 3/32 inch and the overall thickness of the mat shall be not less than 1/2 inch.

3.8 Weight. Each mat shall weigh not less than 18 pounds.

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3.9 Color. The rubber tile surface shall have a black background with white and green mottling.

3.10 Finish. The rubber tile shall be provided with a durable finish which when tested as specified in 4.4.4 shall show good color-fastness and water-resistant characteristics.

3.11 Identification marking. Each rubber floor mat shall be permanently and legibly marked with the manufacturer's name or registered trademark. Markings secured by adhesives shall not be used.

3.12 Workmanship. Workmanship shall be first class in every respect. The mats shall be free of blisters, cracks, imbedded foreign matter, or other physical defects which detract from their appearance or which may impair their serviceability. Mats shall be free of objectionable odors.

4. SAMPLING, INSPECTION AND TEST PROCEDURES.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual, investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2 Certificates of quality. When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105.

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105.

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Characteristic	Requirement	Test procedure	Inspection level *
Flexibility	3.4	4.4.1	S-2
Friction	3.5	4.4.2	S-2
Modulus at 10% elongation	3.6	4.4.3	S-2
Dimensions	3.7	4.3.2	S-2
Colorfastness and water-resistance	3.9	4.4.4	S-2

* Acceptance number shall be zero and not less than 3 samples shall be taken regardless of lot size.

4.3 Examination. The mats shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with the following classification of defects:

Categories and defects *

Major

101. Any area not securely cemented or vulcanized to hold the two layers together.
102. Any surface or edge not free of cracks, tears, blisters, or imbedded foreign matter.
103. Any edge not uniformly beveled as specified.

Minor

201. Identification marking not permanent or legible.
202. Any mat not free of grease, oil or otherwise not clean.

* Examination shall not be restricted to the classified possible defects listed above.

4.3.2 Dimensional examination. The floor mats shall be examined for defects in dimensions. Any dimension not within the tolerance specified shall be classified as a defect.

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4.4 Tests.

4.4.1 Flexibility. The mat shall be tested to determine flexibility characteristics by doubling and pressing the mat flat upon itself for five minutes with the tile surface on the outside. This test shall be applied before and after the mat is subjected to a temperature of 158° F. for 7 days.

4.4.2 Friction. The mats shall be tested to determine compliance with the separation characteristics as specified in Method 8021 of Fed. Test Method Std. 601 using a three pound weight. The specimen tested shall be a strip specimen.

4.4.3 Modulus. The mat shall be tested to determine compliance with the requirement for the modulus, or tensile strength, in accordance with Method 4131 of Fed. Test Method Std. 601.

4.4.4 Water-resistant finish. A small portion of the surface shall be thoroughly cleaned by wiping vigorously with a clean dry cloth. Using a medicine dropper or glass rod, one drop of water shall be placed on the clean surface. After 5 minutes, tap or shake off the water and again wipe off the surface with a clean, dry cloth. The mat shall be unwetted and unmarked and the cloth shall not be discolored by the color of the mat.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. One mat, as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in table I shall be packaged in each unit, intermediate, and exterior container, and as applicable, for the required level of protection specified in the procurement document.

Table I. Packaging Quantities

Unit package	Intermediate package	Exterior container
1 unit	Not applicable	4 units

5.1.2.1 Packing variation permitted. If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

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5.1.3 Level A.

5.1.3.1 Unit package. Each mat shall be completely wrapped in kraft paper, securely taped closed, and further packaged in a fiberboard carton of appropriate size and design conforming to PPP-B-636, carton closure shall be adequate to prevent spilling of contents under normal handling.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for type 2 load, and constructed in accordance with PPP-B-585, style 3, for class 3 use; PPP-B-601, table II, using type III, class 1 plywood; PPP-B-621, table III; or PPP-B-636, type CF, class weather-resistant.

5.2.1.2 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes.

5.2.1.3 Closure. Closure of wood boxes shall be in accordance with appendix of applicable box specification. Closure of each fiberboard box shall be as specified in the appendix, and waterproofing shall conform with 30.4, of PPP-B-636.

5.2.1.4 Strapping. Strapping, when required, shall be in accordance with appendix of applicable box specification.

NOTE: Strapping shall not be required for shipments forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall conform to PPP-B-601, table I, using type III, class 1 plywood; or shall be designed for type 2 load, and constructed in accordance with PPP-B-585, style 3, for class 2 use; PPP-B-621, table I; or PPP-B-636, type CF, class domestic and method I closure shall be utilized.

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5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Interior packages. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Number and date of this specification.
(b) Selection of applicable levels of packing (see 5.1 and 5.2).
(c) When preproduction sample is not required.
(d) Where preproduction sample is to be sent.

6.2 This specification does not cover all types, classes, grades, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet requirements.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Review activities:
Army - MD
Navy - MS
Air Force - 03

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